

CSCE 638 NATURAL LANGUAGE PROCESSING · MODULE 04 · LECTURE NOTES

Word Representations

Meaning from co-occurrence, and the limits of one vector.

MODULE OUTCOMES

- State the distributional hypothesis and its scope.
- Explain how the embedding objectives work.
- Explain what the geometry does and does not encode.
- Explain why one vector per word is insufficient.
- Train and probe embeddings.

1 The distributional hypothesis

WORDS APPEARING IN SIMILAR CONTEXTS TEND TO HAVE SIMILAR MEANINGS

It is an empirical claim rather than a definition — Firth's formulation, 'you shall know a word by the company it keeps' — and **it is true enough to be extremely useful**, which is the only standard that matters for this purpose.

So it converts an apparently semantic problem into a counting problem: represent a word by the distribution of contexts it occurs in, and similar words automatically receive similar representations. No labels, no lexicon, no annotation.

Which solves Module 03 §2's generalisation failure — 'cat' and 'dog' now genuinely *are* close in the representation, so **evidence about one transfers to the other** — and it does so from raw text alone, which is Module 01 §3's argument delivering its first concrete result.

And the limitation is hiding in the word 'tend'. Antonyms occur in near-identical contexts — anything you can say is hot you can say is cold — so **'hot' and 'cold' come out as close neighbours**, which is **entirely correct distributionally and badly wrong semantically**, and is §3's problem.

COUNT AND FACTORISE

build a word x context count matrix, reweight it (positive pointwise mutual information), and reduce the dimension (SVD). Classical, interpretable, and it works well.

PREDICT (word2vec)

skip-gram: given a word, predict its context
 CBOW: given the context, predict the word
 trained with negative sampling, so the expensive softmax over the vocabulary is avoided entirely

GLOBAL + LOCAL (GloVe)

fit the vectors so that their dot product predicts the log co-occurrence count directly

AND THEY ARE CLOSELY RELATED: skip-gram with negative sampling is implicitly factorising a shifted PMI matrix. The prediction framing and the counting framing are the same thing viewed differently.

The equivalence result matters more than it is usually given credit for. Levy and Goldberg showed that skip-gram with negative sampling is implicitly factorising a shifted pointwise mutual information matrix — which means **the neural framing was a better *optimiser* rather than a different *idea***. That is a useful corrective to the narrative that neural methods replaced counting: they computed the same thing more scalably, and the distributional hypothesis was doing the work in both.

2 The geometry

- **Nearest neighbours are genuinely meaningful** — the neighbours of 'Paris' are other cities, the neighbours of 'November' are other months — which is **a real and useful result obtained from counting alone**, and was surprising when first demonstrated.
- **Directions encode relations, approximately** — the vector offset between a country and its capital is roughly consistent across many such pairs, which means the space has learned something structural rather than only similarity.
- **And the analogy arithmetic works, with a caveat that matters: the standard evaluation excludes the three input words from the candidate answers, and without that exclusion a large share of analogies return one of the inputs** rather than the intended answer.
- **So the structure is real and considerably weaker than the famous demonstrations suggest** — which is worth knowing before building anything on vector arithmetic, and **the excluded-inputs caveat is the honest footnote on the most-cited result in the area** and is routinely omitted.
- **And the space has no notion of negation, quantification, or composition** — **it is a similarity space, not a semantics**. There is no vector operation corresponding to 'not', and averaging the words of a sentence loses the word order entirely, which is Module 01 §1's compositionality row still unaddressed.

3 What it gets wrong

ONE VECTOR PER WORD TYPE CANNOT REPRESENT A WORD WITH TWO MEANINGS

'Bank' has a single vector, which must serve both the financial sense and the riverside sense — so it ends up somewhere between the two, **representing neither one well** and being a poor neighbour for both 'money' and 'river'.

Which is Module 01 §2's lexical ambiguity, entirely unresolved — and **the resolution requires the representation to depend on the sentence** rather than only on the word identity, which no static method can provide.

And the same applies to any word whose meaning shifts by domain — 'cell', 'charge', 'model', 'bias', 'significant' — **which is most of the interesting vocabulary in any technical field**, and is why domain adaptation mattered so much in the static era.

So this is the limitation that forced contextual representations (Module 07) — and crucially, **it is a limitation of the *object* rather than of the training method**, so **no better objective fixes it**. You have to change what you are learning, not how you learn it.

- **Antonyms are distributionally similar.** 'Hot' and 'cold' appear in nearly identical contexts, so they are near neighbours — **which is catastrophic for sentiment analysis and for entailment**, where the distinction is the entire task.
- **And the embeddings encode the corpus's social biases, measurably** — occupational, gendered, and racial associations appear directly in the geometry, and can be quantified with the same analogy arithmetic that produces the celebrated examples (Module 12 §2).
- **Which is not a bug in the method: the method faithfully represents the distribution it was given**, and the distribution contains the bias because the text does. **That sentence is the one to remember** — it applies to every model in this course, not only to embeddings.
- **So debiasing the vectors treats a symptom** — and **the measured result is that the standard debiasing methods hide the association rather than removing it**: the clusters remain recoverable, and downstream behaviour changes less than the metric suggests.
- **Which is a general warning about post-hoc corrections**, and one worth carrying into Module 12: a correction that improves the measurement of a problem without improving the problem is **CSCE 701 Module 12's Goodhart failure**, in a research setting.

4 Where static embeddings still earn their place

- **When you need speed.** A lookup table is approximately free, and a contextual model is not — which matters enormously at retrieval scale, where you may be embedding millions of documents (CSCE 670 Module 08).

- **When you have very little data. Pretrained embeddings plus a simple classifier is a strong and extremely cheap baseline** — and **Module 10 §2 requires you to beat it**, which a surprising number of published systems do not do convincingly.
- **For analysis. Measuring semantic change over historical time, or quantifying bias in a corpus, is much easier with exactly one vector per word** than with a contextual model where there is no single vector to examine — so the limitation becomes an advantage.
- **And for initialisation**, which was their original role in neural pipelines and is now largely superseded by pretraining the whole model (Module 07).
- **But for anything compositional or ambiguous, use contextual representations** (Module 07) — which is most real tasks. **The baseline role is the one to remember:** embeddings plus logistic regression takes an hour, costs nothing, and is the number your expensive system has to beat before it has demonstrated anything.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Jurafsky & Martin, chapter 6 (free)	The whole module , including the PPMI weighting and the evaluation caveats of §2.
Mikolov et al., and Pennington et al. — word2vec and GloVe (free)	§1's objectives in the originals — both short, and worth reading for how modest the claims were.
Levy & Goldberg — Neural Word Embedding as Implicit Matrix Factorization (free)	§1's equivalence result — the paper that connected the two framings.
Bolukbasi et al., and Gonen & Goldberg (free)	§3's bias finding, and then the result that debiasing hides rather than removes it — read both, in that order.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the distributional hypothesis** and say what kind of claim it is.
- 2 **Build a word-context count matrix**, apply PPMI, and reduce it with SVD.
- 3 **Train skip-gram with negative sampling** on the same corpus and compare the neighbours.
- 4 **Inspect the nearest neighbours** of ten words and report which are meaningful.
- 5 **Run the analogy arithmetic with and without excluding the inputs**, and report the difference.

- 6 **Find the vector for an ambiguous word** and show it is a poor neighbour for both senses.
- 7 **Measure the similarity of five antonym pairs** and explain the result.
- 8 **Measure one occupational association** in your embeddings.
- 9 **Apply a debiasing method** and check whether the clusters are still recoverable.
- 10 **Build the embeddings-plus-logistic-regression baseline** for your Project 2 task, and record the score.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the distributional hypothesis and where the limitation hides.
- 2 How does it solve Module 03's generalisation failure?
- 3 Name three ways to obtain embeddings and state the equivalence result.
- 4 Why does the equivalence matter?
- 5 What does the geometry encode, and what is the analogy caveat?
- 6 What does the space have no notion of?
- 7 Why can one vector not represent two senses, and why is no better objective enough?
- 8 Why are antonyms similar, and what does that break?
- 9 Why is encoded bias not a bug in the method?
- 10 Give four places static embeddings still earn their place.